DATABASE MANAGEMENT SYSTEMS

ICT

MODULE 4

CAT 3

Candidate Name: _____

Admission Number: _____ Date: _____

Time: 1 hour | **Total:** 30 marks

Instructions

1. Answer ALL questions.
2. Write your name and admission number in the spaces provided above.
3. Show all working and write your answers neatly.

QUESTIONS

1. Define **normalization** and state its main purpose. (2 marks)
2. Briefly explain the three types of anomalies caused by poorly designed tables. (2 marks)
3. Differentiate between **CHAR** and **VARCHAR**, and state which data type you would use for storing prices and why. (2 marks)
4. State **two advantages** and **two disadvantages** of indexing. (2 marks)

5. Distinguish between a **PRIMARY KEY** and a **UNIQUE** constraint. (2 marks)
6. A sports club keeps the following table:

   CLUB_MEMBER

MemberID	**MemberName**	**Sports**
1	Mary	Football, Netball
2	John	Chess
3	Peter	Football, Chess, Swimming

   a) Explain why this table is not in 1NF. (1 mark)

   b) Convert the table into 1NF and state its primary key. (3 marks)

7. Consider this table, which is in 1NF, with composite primary key {OrderID, ProductID}:

   ORDER_ITEM

OrderID	**ProductID**	**ProductName**	**UnitPrice**	**Quantity**
O1	P1	Laptop	60000	2
O1	P2	Mouse	800	5
O2	P1	Laptop	60000	1

a) List the functional dependencies in the table and identify any partial dependency. (2 marks)

b) Decompose the table into 2NF. (2 marks)

c) Give **one** anomaly that the original table suffers from. (1 mark)

8. a) Write an SQL CREATE TABLE statement for **ENROLLMENT** with these attributes:
    - StudentID (integer)
    - CourseID (up to 6 characters)
    - EnrollDate (date, defaulting to the current date)
    - Marks (must be between 0 and 100)

    The primary key is {StudentID, CourseID}. StudentID references STUDENT(StudentID) and CourseID references COURSE(CourseID). (3 marks)

    b) Write a CREATE INDEX statement to index the StudentName column of the STUDENT table, and give one reason why this column is a good candidate for indexing. (2 marks)

9. Use the tables below to answer this question.

    STUDENT(StudentID, StudentName, Department)
    ENROLLMENT(StudentID, CourseID, EnrollDate, Marks)

    a) Write a SELECT query to display the names and departments of all students in the 'ICT' department, arranged alphabetically by name. (2 marks)

    b) Write a query to display the average marks for each CourseID, showing

only courses whose average is above 60. (2 marks)

c) Write a query that joins the two tables to display each student's name, CourseID and Marks. (2 marks)